

Logic Fundamentals

Statements

Logic

- A **statement** or a **claim** is a declarative sentence.
- **There is a police car outside.**
- **Emissions of carbon monoxide into the air have increased in the last ten years.**
- **Vitamin C increases one's resistance to colds.**

A factual statement is an assertion that something ***is or isn't*** a case. We will include in this category empirical claims and conceptual claims.

Examples:

- **There are no four-sided triangles.**
- **Mammals are animals.**
- **Some brothers have no siblings.**

A normative statement is an assertion that something ***ought or ought not*** to be the case.

- **The presence of the police car outside is alarming.**
- **Increase in emissions of carbon monoxide into the air is harmful.**
- **You ought to eat more citrus fruit.**

Descriptive and prescriptive theories

- A **descriptive theory** sets out to describe, explain, and predict certain phenomena. It operates with factual statements.
- A **prescriptive theory** sets out to prescribe judgements, choices, actions, procedures, etc. it operates with factual and with normative statements.

Arguments

Argument: Set of claims, one of which is the conclusion and the rest are premises.

Are its premises true and worthy of our belief?

Does its conclusion really follow from the premises?

Most of the arguments philosophers concern themselves with are **deductive** arguments. Most of the arguments we employ in everyday life are **inductive** arguments.

Valid Arguments

We call an argument **deductively valid** (or, for short, just "valid") when the truth of the premises guarantees the truth of the conclusion.

The following is a valid argument:

- 1. All cats are reptiles.
- 2. Bugs Bunny is a cat.
- 3. Bugs Bunny is a reptile.

- The following is an invalid argument:
- 1. If there is a hedgehog in my gas tank, then my car will not start.
2. My car will not start.
3. Hence, there must be a hedgehog in my gas tank.

Normative arguments

Normative arguments are arguments that employ one or more normative claims, as their premises.

1. If you want to find a good job, you should be diligent.
2. You do want to find a good job.
3. You should be diligent.